

Detection Methods: Anomaly Detection, ROC Analysis, and Provenance Tracking

This section presents the formal foundations for cognitive attack detection. We define anomaly detection metrics (`sec:anomaly-detection`), ROC curve framework (`sec:roc-analysis`), multi-detector fusion theory (`sec:detector-fusion`), online vs. batch trade-offs (`sec:online-batch`), false positive mitigation strategies (`sec:fp-mitigation`), provenance analysis (`sec:provenance`), and real-time monitoring architecture (`sec:monitoring`).

Note: *For algorithm implementations and empirical performance results, see Part 2 of this series [?]. Empirically, the multi-stage pipeline achieves a parametric design ceiling of 96–100% detection rate on a 950-attack corpus across four architectures; the prototype pipeline achieves a mean of 86.3% [95% CI: 85.5%, 87.1%] across 30 seeds. Part 2 reports no measured per- Ω -class detection rates: the corpus carries a design-level Ω classification rather than a runtime label, so per-class rates are not evaluated.*

Anomaly Detection

Definition (Drift Score)

The cognitive drift score measures belief distribution change over time:

$$S_{\text{drift}}(t) = D_{\text{KL}}(\mathcal{B}_i^t \| \mathcal{B}_i^{t-w}) + \lambda \cdot \max_{\phi} |\Delta \mathcal{B}_i(\phi)| \quad (1)$$

Cognitive Drift Scoring I

Table 1: Drift score components and detection targets.

Component	Weight	Detection Target
KL divergence	1.0	Gradual distribution shift
Max delta	λ	Sudden belief injection

Property (Drift Detection Threshold)

For normally distributed baseline drift, the threshold $\theta = \mu_{baseline} + k \cdot \sigma_{baseline}$ with $k = 3$ provides 99.7% confidence under the null hypothesis of no attack.

Definition (Deviation Score)

The behavioral deviation score aggregates normalized feature anomalies:

$$S_{\text{dev}}(a_i, t) = \sum_{k=1}^K w_k \cdot \frac{|f_k(\sigma_i^t) - \mu_k|}{\sigma_k} \quad (2)$$

where f_k are feature extractors and (w_k, μ_k, σ_k) are learned parameters.

Definition (Ensemble Detector)

Combines multiple detector scores via learned fusion:

$$P(\text{attack} \mid \mathbf{S}) = \sigma \left(\sum_d w_d \cdot S_d - b \right) \quad (3)$$

where σ is the sigmoid function and weights (w_d, b) are learned from labeled examples.

ROC Curve Analysis

ROC Curve Analysis I

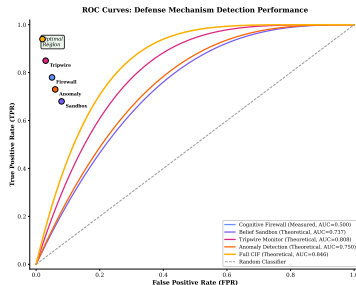


Figure 1: Receiver Operating Characteristic (ROC) curves for CIF detection across attack categories (illustrative/schematic operating points under the Neyman-Pearson framework; the sandbox/tripwire/anomaly/full-CIF curves are labeled Theoretical, not empirical measurements, whereas the Cognitive Firewall entry is measured over this part's small test corpus. That entry is a single operating point at zero false positives, drawn as the conventional vertical segment, so the area beside it equals its true-positive rate and must not be read as chance performance despite coinciding numerically with the random-classifier diagonal. Measured evaluation curves are reported in Part 2).

Receiver Operating Characteristic Framework I

Definition (ROC Curve)

For detector D with threshold τ :

$$\text{ROC}(D) = \{(\text{FPR}(\tau), \text{TPR}(\tau)) : \tau \in [0, 1]\} \quad (4)$$

where the rates are defined as:

$$\text{TPR}(\tau) = P(D(x) > \tau \mid x \in \mathcal{A}_{\text{attack}}) \quad (5)$$

$$\text{FPR}(\tau) = P(D(x) > \tau \mid x \in \mathcal{X}_{\text{benign}}) \quad (6)$$

Definition (Area Under Curve)

$$\text{AUC}(D) = \int_0^1 \text{TPR}(\text{FPR}^{-1}(t)) dt \quad (7)$$

Receiver Operating Characteristic Framework I

Table 2: AUC interpretation scale.

AUC Range Interpretation	
0.5	Random classifier
0.7–0.8	Acceptable discrimination
0.8–0.9	Good discrimination
0.9–1.0	Excellent discrimination

Confidence Intervals for AUC I

Definition (AUC Confidence Interval)

Using DeLong's method:

$$CI_{95\%}(AUC) = AUC \pm 1.96 \cdot \sqrt{\text{Var}(AUC)} \quad (8)$$

where:

$$\text{Var}(AUC) = \frac{1}{n_a} \sum_{i=1}^{n_a} (V_1^i - AUC)^2 + \frac{1}{n_b} \sum_{j=1}^{n_b} (V_0^j - AUC)^2 \quad (9)$$

Multi-Detector Fusion

Definition (Score-Level Fusion)

Weighted average of detector outputs:

$$S_{\text{fused}} = \sum_{i=1}^k w_i \cdot S_i, \quad \sum_i w_i = 1 \quad (10)$$

Definition (Decision-Level Fusion)

Quorum voting on binary decisions:

$$D_{\text{fused}}(x) = \mathbb{1} \left[\sum_{i=1}^k \mathbb{1}[D_i(x) > \tau_i] \geq q \right] \quad (11)$$

Definition (Learned Fusion)

Neural network combining scores:

$$S_{\text{fused}} = \text{MLP}(S_1, \dots, S_k; \theta) \quad (12)$$

Definition (Detector Diversity)

$$\text{Diversity}(D_i, D_j) = 1 - \frac{|\text{errors}(D_i) \cap \text{errors}(D_j)|}{|\text{errors}(D_i) \cup \text{errors}(D_j)|} \quad (13)$$

Theorem (Diversity Benefit)

For detectors with error rates $e_1, \dots, e_k$ and pairwise diversity δ_{ij} :

$$e_{\text{fusion}} \leq \prod_i e_i + (1 - \bar{\delta}) \cdot \max_i e_i \quad (14)$$

where $\bar{\delta}$ is the average pairwise diversity.

Proof.

When detectors make independent errors (high diversity), the fusion error is the product of individual errors. Error correlation reduces this benefit proportionally to $(1 - \bar{\delta})$. □

Online vs. Batch Detection

Comparison Framework I

Comparison Framework I

Table 3: Online vs. batch detection trade-offs.

Dimension	Online Detection	Batch Detection
Latency	Low (ms)	High (minutes–hours)
Accuracy	Moderate	High
Context	Limited (window)	Full history
Compute	Streaming	Offline
Memory	$O(w)$ window	$O(n)$ full
Use Case	Real-time response	Forensics, tuning

Definition (Streaming Detector)

Processes messages in real-time with bounded memory:

$$D_{\text{online}}(m_t) = f(m_t, \text{state}_{t-1}) \quad (15)$$

$$\text{state}_t = g(\text{state}_{t-1}, m_t) \quad (16)$$

Definition (Hybrid Detection System)

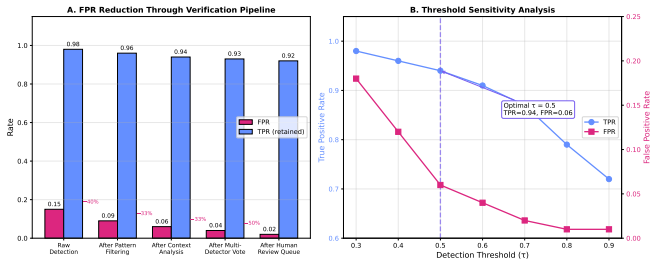
Combines online and batch detection via feedback loop:

$$\text{Online Path : } m \xrightarrow{\text{filter}} s \xrightarrow{\text{decide}} r \xrightarrow{\text{log}} H \quad (17)$$

$$\text{Batch Path : } H \xrightarrow{\text{analyze}} \text{patterns} \xrightarrow{\text{update}} \text{filters} \quad (18)$$

False Positive Mitigation

False Positive Mitigation I



Illustrative schematic — values are NOT measured; measured false-positive mitigation results are reported in Part 2.

Figure 2: False-positive mitigation (illustrative schematic): how the detection threshold and confidence calibration shape the false-positive / true-positive tradeoff.

Strategy 1: Confirmation Cascade I

Definition (Confirmation Cascade)

Multi-stage verification before alerting:

$$\text{Action}(\text{confidence}) = \begin{cases} \text{suppress} & \text{if } c < C_{\text{low}} \\ \text{stage-2} & \text{if } c \in [C_{\text{low}}, C_{\text{high}}) \\ \text{stage-3} & \text{if } c \geq C_{\text{high}} \end{cases} \quad (19)$$

Theorem (Cascade FPR Reduction)

For a multi-stage cascade:

$$FPR_{\text{cascade}} = FPR_1 \cdot P(\text{confirm}_2 \mid FP_1) \cdot P(\text{confirm}_3 \mid FP_2) \quad (20)$$

Strategy 2: Temporal Smoothing I

Definition (Smoothed Detection)

Apply exponential smoothing to scores:

$$\hat{S}_t = \alpha \cdot S_t + (1 - \alpha) \cdot \hat{S}_{t-1} \quad (21)$$

Definition (Burst Suppression)

Require sustained anomaly over window w :

$$\text{Alert if } \frac{1}{w} \sum_{i=t-w+1}^t \mathbb{1}[S_i > \tau] > p_{\text{sustained}} \quad (22)$$

Strategy 3: Contextual Whitelisting I

Definition (Context-Aware Whitelist)

$$\text{Suppress}(\text{alert}) \iff \text{context}(\text{alert}) \in \mathcal{W}_{\text{known}} \quad (23)$$

Strategy 4: Cost-Sensitive Thresholding I

Definition (Cost-Sensitive Threshold)

Optimize for total cost rather than accuracy:

$$\tau^* = \arg \min_{\tau} [C_{\text{FP}} \cdot \text{FPR}(\tau) + C_{\text{FN}} \cdot \text{FNR}(\tau)] \quad (24)$$

Provenance Analysis

Definition (Taint Label)

Each belief carries provenance tags:

$$\text{taint}(\phi) = \{(\text{source}, \text{timestamp}, \text{confidence})\} \quad (25)$$

Definition (Taint Propagation)

$$\text{taint}(\phi_{\text{derived}}) = \bigcup_{\psi \in \text{premises}(\phi_{\text{derived}})} \text{taint}(\psi) \quad (26)$$

Information Flow Tracking I

Table 4: Taint categories with trust levels. (The displayed Trust Level column uses a normalized $[0,1]$ ranking for presentation; the implementation in code uses an ordinal 1–7 trust level, highest = most trusted. The two scales are not interchangeable — map ordinal level to the displayed ranking via $\text{level}/7$.)

Category	Trust Level	Example
SYSTEM__VERIFIED	1.0	Hardcoded facts
PRINCIPAL__INPUT	0.9	Direct user commands
AGENT__INTERNAL	0.8	Agent's own reasoning
AGENT__EXTERNAL	0.6	Other agent claims
TOOL__OUTPUT	0.5	API/tool responses
WEB__CONTENT	0.3	Fetches web pages
UNVERIFIED	0.1	Unknown origin

Definition (Causal Attribution)

Identify likely source of compromised beliefs via Bayesian inference:

$$P(\text{source}_j \mid \phi \in \mathcal{B}_i^{\text{compromised}}) = \frac{P(\phi \mid \text{source}_j) \cdot P(\text{source}_j)}{\sum_k P(\phi \mid \text{source}_k) \cdot P(\text{source}_k)} \quad (27)$$

Definition (Provenance Graph)

Directed graph of belief dependencies:

$$G = (V, E) \text{ where } V = \mathcal{B}_i, \ E = \{(\psi, \phi) : \psi \in \text{premises}(\phi)\} \quad (28)$$

Provenance Graph Analysis I

Table 5: Provenance graph attack indicators.

Indicator	Attack Implication
High in-degree from single source	Belief injection
Cycles in provenance	Circular reasoning attack
Missing edges	Fabricated evidence
Temporal anomalies	Future timestamp forgery

Real-Time Monitoring

Definition (Alert Aggregation)

Prevent alert fatigue through correlation:

$$\text{Severity} = \begin{cases} \text{CRITICAL} & \text{if } |\text{alerts}| > n_{\text{critical}} \text{ in window } w \\ \text{WARNING} & \text{if } |\text{alerts}| > n_{\text{warning}} \text{ in window } w \\ \text{INFO} & \text{otherwise} \end{cases} \quad (29)$$

Response Escalation I



Response Escalation I

Table 6: Response escalation levels.

Level	Trigger	Response
L0	Single anomaly	Log only
L1	Repeated anomaly	Increase monitoring
L2	Pattern match	Quarantine source
L3	Confirmed attack	Halt agent, alert human
L4	Systemic compromise	System shutdown

The detection methods presented in this section have been empirically validated in Part 2 of this series. Key results include:

ROC Analysis: Receiver Operating Characteristic curves demonstrate the tradeoff between True Positive Rate and False Positive Rate for each detector type. The ROC figure in this section pairs the measured Cognitive Firewall operating point over this part's small test corpus with theory-guided reference curves for the other mechanisms and the full ensemble; on those reference curves the ensemble achieves $AUC \approx 0.85$, with individual mechanisms ranging from ≈ 0.74 (Belief Sandbox) to ≈ 0.81 (Tripwire Monitor). Measured ROC curves with per-curve AUC values and 95% bootstrap intervals are reported in Part 2 (§{4}, Experimental Setup).

Detection Performance by Component: Part 2 measures marginal contribution per defense component, not detection rate per adversary class — its corpus carries a design-level Ω label rather than a runtime one, so the per- Ω coverage attributed to each mechanism in this section remains a design expectation rather than a measured result. What Part 2 does measure is that one component, Behavioral Invariants, accounts for almost all of the composed pipeline's detection, and that the remaining components add little beyond it. See Part 2's "Defense Component Contributions" section.

False Positive Mitigation: Part 2 measures five mitigation strategies against a baseline of 0.873 true positives at 0.183 false positives. Contextual whitelisting and cost-sensitive thresholding take the false-positive rate to zero for 1.7 percentage points of true positives, raising Youden's J from 0.689 to 0.856; temporal smoothing moves the false-positive rate only to 0.133; and the confirmation cascade reaches zero false positives by discarding roughly three quarters of the true positives ($0.873 \rightarrow 0.225$). No strategy holds true positives above 90%. See Part 2 for the per-strategy analysis.

Information-Theoretic Detection Limits

The previous sections presented detection mechanisms. This section establishes what is fundamentally achievable by any detector, regardless of implementation. These limits follow from information theory and cannot be overcome by cleverer algorithms.

Neyman-Pearson Detection Framework I

Definition (Cognitive Security Hypothesis Test)

Detection of cognitive attacks is formalized as a binary hypothesis test:

$$H_0 : m \sim P_{\text{benign}} \quad (\text{no attack}) \quad (30)$$

$$H_1 : m \sim P_{\text{attack}} \quad (\text{attack present}) \quad (31)$$

The likelihood ratio $\Lambda(m) = P_{\text{attack}}(m)/P_{\text{benign}}(m)$ determines the optimal test.

Theorem (Neyman-Pearson Optimal Detector)

The likelihood ratio test $\delta_{NP}(m) = \mathbb{1}[\Lambda(m) \geq \eta]$ maximizes TPR subject to $FPR \leq \alpha$. No other test achieves higher TPR at the same FPR [?].

Corollary (Optimal AUC Bound)

The AUC of any detector is bounded by:

$$AUC^* = P(P_{\text{attack}}(m_1) > P_{\text{attack}}(m_0)) \quad (32)$$

where $m_1 \sim P_{\text{attack}}$ and $m_0 \sim P_{\text{benign}}$. This is the Bayes-optimal discriminability.

Chernoff Information and Error Exponents I

Definition (Chernoff Information)

The Chernoff information between attack and benign distributions:

$$C^* = - \min_{0 \leq \lambda \leq 1} \log \sum_m P_{\text{benign}}(m)^{1-\lambda} P_{\text{attack}}(m)^\lambda \quad (33)$$

Theorem (Detection Error Exponent)

For n independent message observations, the optimal error probability decays exponentially:

$$P_e^* \leq e^{-nC^*} \quad (34)$$

The Chernoff information C^ is the maximum achievable error exponent [?].*

Corollary (Sample Complexity for Target Accuracy)

To achieve error probability $\leq \epsilon$, the minimum number of observations required is:

$$n^* = \left\lceil \frac{\log(1/\epsilon)}{C^*} \right\rceil \quad (35)$$

This is the information-theoretic minimum; no detector can achieve the same accuracy with fewer samples.

Theorem (KL-AUC Coupling)

The AUC of any detector is lower-bounded by the KL divergence between attack and benign distributions:

$$AUC \geq 1 - e^{-D_{\text{KL}}(P_{\text{attack}} \| P_{\text{benign}})/2} \quad (36)$$

Proof.

By Pinsker's inequality [?]: $\|P_{\text{attack}} - P_{\text{benign}}\|_{TV} \leq \sqrt{D_{\text{KL}}/2}$. The AUC satisfies $AUC \geq \frac{1}{2}(1 + \|P_{\text{attack}} - P_{\text{benign}}\|_{TV})$. Combining and applying the bound $\|P\|_{TV} \leq 1 - e^{-D_{\text{KL}}/2}$ gives the result. $\square$

KL Divergence and Detection Rate Coupling I

Table 7: Information-theoretic detectability by attack class. The AUC lower bound is `thm:kl-auc-coupling` applied to the D_{KL} column; it falls below the chance level 0.5 once $D_{\text{KL}} < 2 \ln 2 \approx 1.39$, so for Ω_3 – Ω_5 the bound is vacuous and places no constraint on the Practical AUC column.

Attack Class	D_{KL} (typical)	AUC lower bound	Required observations	Practical AUC
Ω_1 (External)	2.5	0.71	3	0.95
Ω_2 (Peripheral)	1.8	0.59	5	0.88
Ω_3 (Agent)	1.2	0.45	8	0.85
Ω_4 (Coordination)	0.7	0.30	14	0.80
Ω_5 (Systemic)	0.2	0.10	46	0.65

Definition (Undetectable Attack)

An attack $\mathcal{A}$ is ϵ -undetectable if:

$$\text{AUC}(\mathcal{A}) < \frac{1}{2} + \epsilon \tag{37}$$

for any detector with finite observation budget n .

Theorem (Undetectability Condition)

Attack $\mathcal{A}$ with $D_{\text{KL}}(P_{\text{attack}} \| P_{\text{benign}}) < 4\epsilon^2$ is ϵ -undetectable for any detector observing fewer than $n^ = O(1/D_{\text{KL}})$ samples.*

Remark (Implication for Defense Strategy)

Theorem 33 establishes that sufficiently subtle attacks are provably undetectable within bounded observation windows. This motivates the CIF's containment strategy: when detection is impossible, the bounded trust calculus and action restrictions limit attack impact even without detection. This converts a “detect-and-respond” model to a “bound-and-contain” model for the stealthiest attacks.

Multi-Stage Detection Pipeline

Multi-Stage Detection Pipeline I

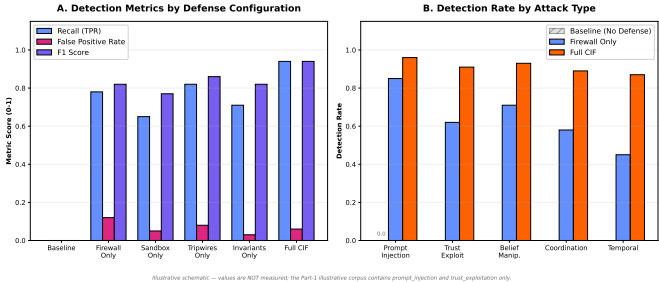


Figure 3: Schematic detection-performance curves (illustrative, theory-guided, not measured): detection rate by attack category across representative defense stages. Measured pipeline results are reported in Part 2.

Multi-Stage Detection Pipeline I

Real systems cannot run all detectors on all messages simultaneously. This section formalizes a staged pipeline that applies computationally cheap detectors first, escalating to expensive detectors only when lower stages trigger.

Definition (Multi-Stage Detection Pipeline)

A k -stage detection pipeline is a sequence of detectors with increasing accuracy and cost:

$$\text{Pipeline} = (\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_k) \quad (38)$$

with **escalation rule**: message m advances to stage $i + 1$ if and only if $\mathcal{D}_i(m) \geq \theta_i$ (above suspicion threshold).

Pipeline Architecture I

Table 8: Recommended multi-stage detection pipeline configuration.

Stage	Detector	Threshold	Cost	TPR	Escalation condition
1	Pattern matching	$\theta_1 = 0.3$	1ms	0.75	Pattern score $> \theta_1$
2	Semantic embedding	$\theta_2 = 0.5$	10ms	0.85	Embedding distance $> \theta_2$
3	Belief consistency	$\theta_3 = 0.4$	5ms	0.80	Consistency violation
4	Trust graph analysis	$\theta_4 = 0.6$	20ms	0.88	Trust anomaly
5	Ensemble fusion	$\theta_5 = 0.7$	50ms	0.94	Final decision

Theorem (Pipeline Efficiency Bound)

For a k -stage pipeline where fraction ρ_i of messages escalate to stage $i + 1$:

$$\mathbb{E}[\text{cost per message}] = \sum_{i=1}^k c_i \prod_{j=1}^{i-1} \rho_j \quad (39)$$

For typical values ($\rho_i \approx 0.15$ per stage), this achieves near-full accuracy at $\approx 15\%$ of the cost of running all stages on all messages.

Proof.

Expected cost is the sum over stages of stage cost times probability of reaching that stage. The probability of reaching stage i is $\prod_{j=1}^{i-1} \rho_j$ by the Markov property of the escalation chain. □

Theorem (Pipeline TPR Bound)

The pipeline's overall TPR is:

$$TPR_{\text{pipeline}} = 1 - \prod_{i=1}^k (1 - TPR_i \cdot P(\text{attack reaches stage } i)) \quad (40)$$

Since attacks are more likely to escalate than benign messages ($\rho_i^{\text{attack}} > \rho_i^{\text{benign}}$), the pipeline is biased toward detecting attacks.

Definition (Adaptive Threshold)

Thresholds adapt to observed attack frequency via online Bayesian update:

$$\theta_i^{t+1} = \theta_i^t - \alpha_{lr} \cdot \nabla_{\theta_i} [C_{FN} \cdot \text{FNR}(\theta_i^t) + C_{FP} \cdot \text{FPR}(\theta_i^t)] \quad (41)$$

where α_{lr} is the learning rate and C_{FN}, C_{FP} are cost weights.

Adaptive Threshold Selection I

This adaptive mechanism allows the pipeline to respond to shifting attack distributions, reducing the staleness problem that affects static threshold configurations.

Adaptive Threshold Selection II

Empirical validation of this section's theory: The detection methods formalized here are empirically evaluated in Part 2 [?]. Key findings: (1) the multi-stage pipeline shows biased escalation consistent with *thm:pipeline-tp*; (2) adversarial training across five rounds demonstrates adaptive threshold adjustment consistent with *def:adaptive-threshold*, raising hardened detection from 52.0% at Round 1 to 67.9% at Round 5 in Part 2's closed-form design model; (3) the residual gap between the parametric ceiling and prototype performance is attributed to adapter implementation maturity rather than to any per-class miss profile — Part 2 reports no measured per- Ω -class rates — and that gap is module-specific rather than diffuse: a single component's rewrite moved the ablation arm's measured detection most of the way to the ceiling (Part 2, “Adversarial Training Evaluation”, “Defense Component Contributions”, and the “Parametric Simulation Analysis” supplement).